Let’s return for a final time to the atoms of Democritus. Let’s imagine a world formed of an interminable sea of atoms which bounce, attract and cling together, and of nothing else. Aren’t we missing something?

Plato and Aristotle insisted on the fact that something was indeed missing; they thought that the *form* of things was this something *extra* that had to be added to the *substance* of which things were made in order to understand the world. For Plato, forms exist by themselves, in an ethereal ideal world of forms, a world of ‘ideas’. The idea of a horse exists prior to and independently of any actual horse. For Plato, a real horse is nothing but a pale reflection of the idea of a horse. The atoms which make up the horse count for little: what counts is the ‘horseness’, the abstract form. Aristotle is a bit more realistic, but for him, too, the form cannot be reduced to the substance. In a statue, there is more than the stone of which it is made. This *more*, for Aristotle, is the form. This is the basis of the critique of Democritus’s materialism in antiquity. It still remains a common critique of materialism.

But was Democritus really proposing that everything can be reduced to atoms? Let’s look at it more closely. Democritus says that when atoms combine what counts is their form, their arrangement in the structure, as well as the way in which they *combine*. He gives the example of the letters of the alphabet: there are only twenty or so letters but, as he puts it, ‘It is possible for them to combine in diverse modes, in order to produce comedies or tragedies, ridiculous stories or epic poems.’

There are more than just atoms in this idea: what counts is the *way* in which they are combined, one in relation to another. But what relevance can the way in which they are combined have, in a world in which there is nothing but other atoms?

If the atoms are also an alphabet, who is able to read the phrases written with this alphabet?

The answer is subtle: the way in which the atoms arrange themselves is correlated with the way *other* atoms arrange themselves. Therefore, a set of atoms can have *information*, in the technical, precise sense described above, about another set of atoms.

This, in the physical world, happens continuously and throughout, in every moment and in every place: the light which arrives at our eyes carries information about the objects which it has played across; the colour of the sea has information on the colour of the sky above it; a cell has information about the virus that is attacking it; a new living being has plenty of information because it is correlated with its parents, and with its species; and you, dear reader, when reading these lines, receive information about what I am thinking while writing them, that is to say, about what is happening in my mind at the moment in which I write this text. What occurs in the atoms of